GOVERNMENT OF KHYBER PAKHTUNKHWA

KHYBER PAKHTUNKHWA TEXTBOOK BOARD PESHAWAR

OFFICIAL PRESCRIBED E-CURRICULUM TEXTBOOK

MATHEMATICS SCIENCE CLASS 10

CLASS / GRADE: 10th (Matric)

Subject: Mathematics | Language: English

AUTHENTIC CURRICULUM SYLLABUS & CHAPTER BLUEPRINTS

· Total Complete Curriculum Chapters: 12 Units

· Designed as per National Curriculum & KPK BISEP Standards

· Contains complete theory, conceptual exercises, MCQs & solutions

· Fully synchronized with ExamForge AI Examination Paper Generator

Published by: Khyber Pakhtunkhwa Textbook Board, Phase-V, Hayatabad, Peshawar

Approved by: Directorate of Curriculum and Teacher Education (DCTE) Abbottabad, KP

Academic Year: 2026 Examination Edition · All Rights Reserved

TABLE OF CONTENTS · MATHEMATICS SCIENCE CLASS 10

Unit 01:	Quadratic Equations: Factorization, Completing Square, Quadratic	Page 3
Unit 02:	Theory of Quadratic Equations: Discriminant, Nature of Roots, Cub	Page 5
Unit 03:	Variations: Direct & Inverse Variations, Joint Variation, Theorem	Page 7
Unit 04:	Partial Fractions: Proper, Improper, Linear & Quadratic Factors	Page 9
Unit 05:	Sets and Functions: Binary Relations, Functions, De Morgan's Laws	Page 11
Unit 06:	Basic Statistics: Frequency Distribution, Mean, Median, Mode, Var	Page 13
Unit 07:	Introduction to Trigonometry: Measurement of Angles, Trigonometri	Page 15
Unit 08:	Projection of a Side of a Triangle & Hero's Formula	Page 17
Unit 09:	Chords of a Circle & Theorems	Page 19
Unit 10:	Tangent to a Circle & Theorems	Page 21
Unit 11:	Chords and Arcs	Page 23
Unit 12:	Practical Geometry - Circles	Page 25

UNIT 1: QUADRATIC EQUATIONS: FACTORIZATION, COMPLETING SQU

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Quadratic Equations: Factorization, Completing Square, Quadratic F

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 2: THEORY OF QUADRATIC EQUATIONS: DISCRIMINANT, NATUR

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Theory of Quadratic Equations: Discriminant, Nature of Roots, Cube

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 3: VARIATIONS: DIRECT & INVERSE VARIATIONS, JOINT VAR

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Variations: Direct & Inverse Variations, Joint Variation, Theorem on

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 4: PARTIAL FRACTIONS: PROPER, IMPROPER, LINEAR & QUAD

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Partial Fractions: Proper, Improper, Linear & Quadratic Factors.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 5: SETS AND FUNCTIONS: BINARY RELATIONS, FUNCTIONS, D

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Sets and Functions: Binary Relations, Functions, De Morgan's Laws

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 6: BASIC STATISTICS: FREQUENCY DISTRIBUTION, MEAN, ME

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Basic Statistics: Frequency Distribution, Mean, Median, Mode, Varia

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 7: INTRODUCTION TO TRIGONOMETRY: MEASUREMENT OF ANGLE

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Introduction to Trigonometry: Measurement of Angles, Trigonometri

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 8: PROJECTION OF A SIDE OF A TRIANGLE & HERO'S FORMUL

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Projection of a Side of a Triangle & Hero's Formula.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 9: CHORDS OF A CIRCLE & THEOREMS

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Chords of a Circle & Theorems.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 10: TANGENT TO A CIRCLE & THEOREMS

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Tangent to a Circle & Theorems.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 11: CHORDS AND ARCS

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Chords and Arcs.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.

UNIT 12: PRACTICAL GEOMETRY - CIRCLES

Class 10th · Mathematics · Khyber Pakhtunkhwa Textbook Board Peshawar

1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Practical Geometry - Circles.

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum. Students are guided through structured topic explanations, standard definitions, solved examples, and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
Teachers can generate Section A, B and C papers with custom marks & choices instantly.